Analysis I

Uebungsblatt 5

Problem:	Comment:	Points:
1		31 / 34

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(a)

Proof. Let

$$a_n := (1 + (-1)^n)\frac{n+1}{n} + (-1)^n.$$

Split by parity:

$$(-1)^n = \begin{cases} 1, & n \text{ even}, \\ -1, & n \text{ odd}. \end{cases}$$

Hence

$$a_n = \begin{cases} 2\frac{n+1}{n} + 1 = 3 + \frac{2}{n}, & n \text{ even}, \\ -1, & n \text{ odd}. \end{cases}$$

Therefore the even subsequence satisfies $a_{2k} = 3 + \frac{1}{k} \to 3$, and the odd subsequence is constant $a_{2k+1} = -1$. Thus the set of accumulation points is $\{-1, 3\}$, realized by the subsequences (a_{2k+1}) and (a_{2k}), respectively. □

(b)

Proof. Let $u_n := \sqrt[n]{n} - 1 > 0$. Since $\sqrt[n]{n} \to 1$, we have $u_n \to 0$. Fix $r \in (0, 1)$, e.g. $r = \frac{1}{2}$. Then there exists N such that $0 < u_n < r$ for all $n \geq N$. Hence, for $n \geq N$,

$$0 \leq (\sqrt[n]{n} - 1)^n = u_n^n \leq r^n \xrightarrow[n\to\infty]{} 0.$$

By the squeeze theorem, $(\sqrt[n]{n} - 1)^n \to 0$. □

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(a)

Proof. Define the sequence (x_n) by

$$x_1 := 7, \qquad x_2 := -5, \qquad x_{2k} := 2 - \frac{1}{k}\ (k \geq 2), \qquad x_{2k+1} := \frac{1}{k}\ (k \geq 1).$$

Then the sequence is bounded with $-5 \leq x_n \leq 7$.

- Supremum and infimum:

$$\sup_{n\in\mathbb{N}} x_n = 7 \quad (\text{attained at } n = 1), \qquad \inf_{n\in\mathbb{N}} x_n = -5 \quad (\text{attained at } n = 2).$$

- Limsup: For even indices with $k \geq 2$ we have $x_{2k} = 2 - \frac{1}{k}$ and hence $x_{2k} \to 2$ with $x_{2k} < 2$ for all k. All odd terms satisfy $x_{2k+1} \leq 1$. Therefore, for every n,

$$\sup_{m\geq n} x_m \leq 2.$$

Given $\varepsilon > 0$, choose K so that $\frac{1}{K} < \varepsilon$. If $n \le 2K$ then

$$\sup_{m \ge n} x_m \ge x_{2K} = 2 - \frac{1}{K} > 2 - \varepsilon.$$

Thus $\limsup_{n\to\infty} x_n = 2$.

- Liminf: For $k \ge 1$ we have $x_{2k+1} = \frac{1}{k}$ and hence $x_{2k+1} \to 0$, while $x_m \ge 0$ for all $m \ge 3$. Let $\varepsilon > 0$ and pick K with $\frac{1}{K} < \varepsilon$. For any $n \ge 3$ choose $j \ge \max\{K, \lceil n/2 \rceil\}$ and set $m := 2j + 1 \ge n$. Then

$$x_m = \frac{1}{j} \le \frac{1}{K} < \varepsilon,$$

so $0 \le \inf_{m \ge n} x_m \le \varepsilon$ for all sufficiently large n. Hence $\liminf_{n\to\infty} x_n = 0$.

All four numbers are finite and pairwise distinct:

$$\inf x_n = -5, \qquad \liminf x_n = 0, \qquad \limsup x_n = 2, \qquad \sup x_n = 7.$$

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□

(b)

Proof. Let (x_n) and (y_n) be bounded sequences. For each $N \in \mathbb{N}$ define the tail suprema

$$s_N := \sup\{x_n : n \ge N\}, \qquad t_N := \sup\{y_n : n \ge N\}, \qquad u_N := \sup\{x_n + y_n : n \ge N\}.$$

By definition of limsup for bounded sequences,

$$\limsup_{n\to\infty} x_n = \lim_{N\to\infty} s_N, \qquad \limsup_{n\to\infty} y_n = \lim_{N\to\infty} t_N, \qquad \limsup_{n\to\infty}(x_n + y_n) = \lim_{N\to\infty} u_N.$$

Fix N. For every $n \ge N$ we have $x_n \le s_N$ and $y_n \le t_N$, hence

$$x_n + y_n \le s_N + t_N \qquad \text{for all } n \ge N.$$

Taking the supremum over all $n \ge N$ yields

$$u_N = \sup\{x_n + y_n : n \ge N\} \le s_N + t_N \qquad \text{for every } N.$$

Now pass to the limit $N \to \infty$:

$$\limsup_{n\to\infty}(x_n+y_n) = \lim_{N\to\infty} u_N \le \lim_{N\to\infty}(s_N+t_N) = \lim_{N\to\infty} s_N + \lim_{N\to\infty} t_N = \limsup_{n\to\infty} x_n + \limsup_{n\to\infty} y_n.$$

This proves

$$\limsup_{n\to\infty}(x_n + y_n) \le \limsup_{n\to\infty} x_n + \limsup_{n\to\infty} y_n.$$

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□

(c)

Proof. Construct bounded sequences (x_n) and (y_n) by

$$x_n = \begin{cases} 1, & n \text{ even}, \\ 0, & n \text{ odd}, \end{cases} \qquad y_n = \begin{cases} 0, & n \text{ even}, \\ 1, & n \text{ odd}. \end{cases}$$

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Then $\limsup x_n = 1$ and $\limsup y_n = 1$. Moreover $x_n + y_n = 1$ for all n, hence

$$\limsup_{n\to\infty}(x_n + y_n) = 1 < 1 + 1 = \limsup_{n\to\infty} x_n + \limsup_{n\to\infty} y_n.$$

□

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Proof. Let $(x_n)_{n\in\mathbb{N}} \subset \mathbb{R}^k$ be bounded, where

$$x_n = (x_{n,1}, \dots, x_{n,k}), \qquad \|x_n\| = \Big(\sum_{l=1}^{k} |x_{n,l}|^2\Big)^{1/2}.$$

Then there exists $M > 0$ such that $\|x_n\| \le M$ for all n, hence for each l,

$$|x_{n,l}| \le \|x_n\| \le M \qquad (n \in \mathbb{N}),$$

so each scalar sequence $(x_{n,l})_{n\in\mathbb{N}}$ is bounded.

Diagonal extraction We build a nested family of index subsequences $(n_{r,j})_{j\in\mathbb{N}}$ for $r = 1, \dots, k$ with ?

I don't understand here

$$n_{1,1} < n_{1,2} < \cdots, \quad n_{2,1} < n_{2,2} < \cdots \quad \text{and} \quad (n_{2,j})_j \subset (n_{1,j})_j, \ \dots, \ (n_{k,j})_j \subset (n_{k-1,j})_j,$$

such that for each $r = 1, \dots, k$ there exists $L_r \in \mathbb{R}$ with

$$x_{n_{r,j},r} \longrightarrow L_r \qquad (j \to \infty),$$

and, because we always pass to subsequences, also

$$x_{n_{r,j},t} \longrightarrow L_t \qquad \text{for every } t = 1, \dots, r.$$

- Step $r = 1$: Apply Bolzano–Weierstrass in $\mathbb{R}$ to $(x_{n,1})_n$ to get a strictly increasing $(n_{1,j})_j$ with $x_{n_{1,j},1} \to L_1$.

- Inductive step: Suppose $(n_{r-1,j})_j$ is chosen with $x_{n_{r-1,j},t} \to L_t$ for $t = 1, \dots, r-1$. The sequence $(x_{n_{r-1,j},r})_j$ is bounded, so by Bolzano–Weierstrass there exists a strictly increasing map $j \mapsto m_j$ such that

$$x_{n_{r-1,m_j},r} \longrightarrow L_r.$$

Define the next index subsequence by $n_{r,j} := n_{r-1,m_j}$. Then $(n_{r,j})_j$ is strictly increasing, $(n_{r,j})_j \subset (n_{r-1,j})_j$, and for $t = 1, \dots, r-1$ we still have $x_{n_{r,j},t} \to L_t$ (subsequence of a convergent sequence), while $x_{n_{r,j},r} \to L_r$ by construction.

After k steps we have $(n_{k,j})_j$ such that, for every $t = 1, \dots, k$,

$$x_{n_{k,j},t} \longrightarrow L_t \qquad (j \to \infty).$$

Let $L := (L_1, \dots, L_k) \in \mathbb{R}^k$.

Vector convergence. Fix $\varepsilon > 0$. For each $t = 1, \dots, k$, choose J_t so that

$$|x_{n_{k,j},t} - L_t| < \frac{\varepsilon}{\sqrt{k}} \qquad \text{for all } j \geq J_t.$$

Let $J := \max\{J_1, \dots, J_k\}$. Then for $j \geq J$,

$$\|x_{n_{k,j}} - L\| = \Big(\sum_{t=1}^{k} |x_{n_{k,j},t} - L_t|^2\Big)^{1/2} \leq \Big(\sum_{t=1}^{k} \frac{\varepsilon^2}{k}\Big)^{1/2} = \varepsilon.$$

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Hence $x_{n_{k,j}} \to L$.

Conclusion. We have produced a convergent subsequence $(x_{n_{k,j}})_j$ of (x_n). Therefore every bounded sequence in $\mathbb{R}^k$ has an accumulation point. □

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Proof. Let $q \in [0, 1)$ and suppose

$$|a_{n+1} - a_n| \leq q\,|a_n - a_{n-1}| \qquad \text{for all } n \geq 2.$$

Fix $n \in \mathbb{N}$ and $p \in \mathbb{N}$. Use the telescoping identity

$$a_{n+p} - a_n = (a_{n+1} - a_n) + (a_{n+2} - a_{n+1}) + \cdots + (a_{n+p} - a_{n+p-1}) = \sum_{j=n}^{n+p-1} (a_{j+1} - a_j),$$

hence, by the triangle inequality,

$$|a_{n+p} - a_n| \leq \sum_{j=n}^{n+p-1} |a_{j+1} - a_j|.$$

For each $j \geq 2$, iterating the recursive bound gives

$$|a_{j+1} - a_j| \leq q\,|a_j - a_{j-1}| \leq q^2\,|a_{j-1} - a_{j-2}| \leq \cdots \leq q^{j-1}\,|a_2 - a_1|.$$

Substituting into the previous estimate yields

$$|a_{n+p} - a_n| \leq |a_2 - a_1| \sum_{j=n}^{n+p-1} q^{j-1} = |a_2 - a_1|\, q^{n-1} \sum_{t=0}^{p-1} q^t = |a_2 - a_1|\, q^{n-1}\, \frac{1 - q^p}{1 - q} \leq \frac{|a_2 - a_1|}{1 - q}\, q^{n-1}.$$

If $q = 0$, then $|a_{n+1} - a_n| = 0$ for all $n \geq 2$, so (a_n) is eventually constant and hence Cauchy. If $0 < q < 1$, let $\varepsilon > 0$ and choose N such that

$$\frac{|a_2 - a_1|}{1 - q}\, q^{N-1} < \varepsilon.$$

For any $m > n \geq N$ (write $m = n + p$ with $p \geq 1$), the bound above gives

$$|a_m - a_n| = |a_{n+p} - a_n| < \varepsilon.$$

Therefore (a_n) is a Cauchy sequence. □

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